

Kubernetes CronJob: The Powerful Job Scheduler

Businesses are always looking for ways to deliver software faster at minimal cost. Adoption of cloud technologies not only makes this easier but also reduces operational cost. Containerisation has emerged as a popular strategy for cloud adoption. This is where Kubernetes CronJob, a powerful tool for scheduling and running periodic jobs in a containerised environment, comes in. Let's see how it works.



Containerisation is a technology that has been gaining popularity in recent years due to its ability to make applications portable, scalable, and easily deployable. Containers provide a lightweight and flexible way to package and distribute software applications; they help to build once and deploy anywhere by encapsulating an application and its dependencies in a container that can run on any host operating system with the same results. Containers are isolated from each other and the host operating system, providing a secure environment for running applications.

Cloud containerisation has further enhanced the advantages of containerisation by enabling users to host their containers on cloud platforms like AWS, Google Cloud Platform, and Microsoft Azure. Container orchestrators such as Kubernetes and cloud-managed implementations like EKS, AKS, GKE, and AWS ROSA automate the deployment as well as auto scaling of containers and their dependencies.

Batch jobs

Batch jobs are computer programs processed in groups, which are used for long-running tasks that can be executed at a later time. They can be made to run on specific schedules such as daily, weekly, or fortnightly, using scheduled tasks in Windows or Cron jobs in Linux and UNIX systems.

While Linux Cron jobs are suitable for basic job scheduling needs, they may not be adequate for more complex or mission-critical workloads. Leveraging this gap, many enterprise job scheduling tools have emerged and gained traction, offering advanced job scheduling features like complex jobs and workflows.

Control-M is a workload automation platform that is being adopted widely as it helps organisations manage their workloads across hybrid environments from a single interface.

Control-M features

Control-M offers many features that make it an ideal choice for managing complex workloads in the cloud.

- **Unified control:** Control-M provides a unified interface for managing workloads across different platforms, including cloud environments like AWS.
- **Advanced scheduling:** It offers advanced job scheduling capabilities that allow users to define complex workflows and dependencies management among jobs with ease.
- **Monitoring and alerting:** Control-M provides real-time monitoring and alerting capabilities that enable users to quickly identify and resolve issues with their workloads.
- **Self-service:** It enables self-service automation for developers, allowing them to easily deploy and manage their applications in the cloud.
- **Control modules (CM):** These integrate with external systems like SAP, Oracle E-Business Suite, etc.

Kubernetes CronJob

Kubernetes provides a powerful and flexible Cron job functionality that enables users to schedule and run batch jobs on a regular basis. These are Kubernetes native objects,